

Closed-loop fiber optic gyros

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INTRODUCTION

Research and development of fiber optic gyros began in the mid 1970s and focused on improving the gyro's sensitivity to rotation and reducing noise. Next, bias performance was addressed. By the early 1980s, fiber gyros were achieving bias errors of 0.01 %/hr in a laboratory environment. Scale factor performance was initially addressed at McDonnell Douglas¹ in the late 1970s by the use of a closed-loop fiber gyro which employed acousto-optic frequency shifters. Good performance was achieved but this approach was not productionized.

In the mid 1980s, Thomson CSF developed a fiber gyro which used a double closed-loop technique employing a digital phase ramp and an electro-optic phase modulator². Derivatives of this approach have been adopted by most gyro producers in the world. This technique has enabled fiber gyros to have high scale factor linearity and has significantly improved scale factor stability and repeatability.

Today closed-loop fiber optic gyros using derivatives of this technique are in production for many tactical applications³. These include tactical missiles, smart bombs, and attitude and heading reference systems (AHRS) and require gyro bias performance of 1 to 10 %/hr and gyro scale factor performance of 100 to 1000 ppm. Closed-loop fiber gyros using derivatives of this technique are presently in development for future inertial navigation systems⁴ (INS) which require bias performance in the 0.001 to 0.01 %/hr and scale factor performance in the 5 to 50 ppm range.

This paper will examine how this double closed-loop, digital phase ramp technique functions. An error source unique to this type of closed-loop gyro, the deadband error, will be examined along with a technique for eliminating it.

CLOSED-LOOP FIBER GYROS

The closed-loop gyro is based on the well-known reciprocal open-loop gyro configuration. The closed-loop gyro adds two components to this configuration. The first component is an optical feedback element which is placed in the fiber sensing loop as shown in figure 1. It is driven by a control unit which accepts an electrical input signal, $V_{c,in}$, and outputs an electrical signal, $S_{c,out}$, to the feedback element. The output signal is linearly related to the input signal via a constant, K_c . The function of this optical feedback unit is to provide a feedback phase shift between the CW and CCW optical waves in the fiber sensing loop proportional to the electrical output signal via a constant K_f .

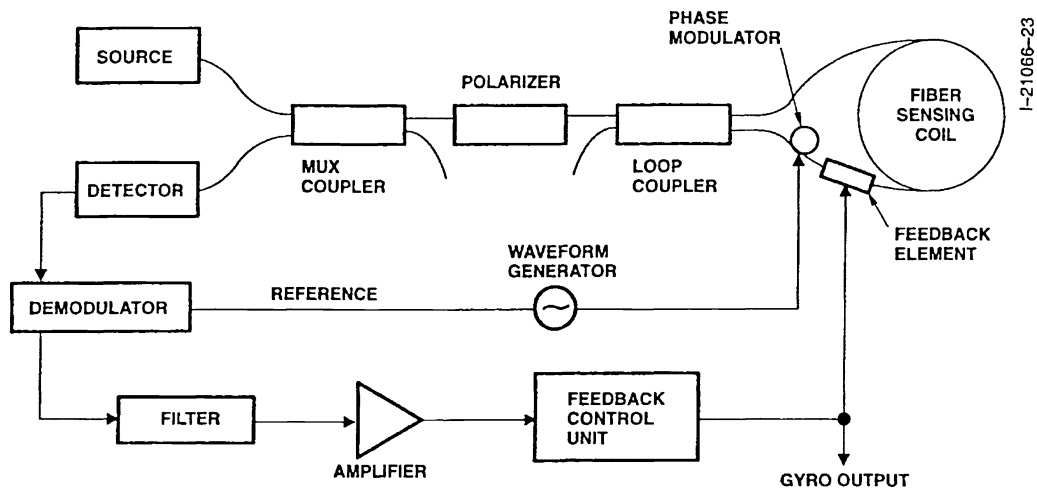


Figure 1. Schematic of a closed-loop fiber gyro.

The second component is a feedback loop which takes the output of the demodulator, filters and amplifies it, and applies the filtered and amplified signal to the input of the optical feedback control unit. If the filtering and amplification are properly adjusted then the feedback loop will adjust the feedback phase shift so that it is always nearly equal and opposite in sign to the Sagnac phase for all constant input rotation rates.

$$\phi_f \approx -\phi_s \quad \text{Eq. (1)}$$

Eq. (1) implies that the total phase shift (i.e. the sum of the feedback and the Sagnac phase shifts) around the fiber sensing loop is nearly equal zero. Such fiber gyros are also referred to as phase nulling gyros. The output of the closed-loop fiber gyro is now the electrical signal applied to the feedback element.

Eq. (1) holds strictly for steady-state conditions. For dynamic environments (i.e. where the angular acceleration is non-zero) the relationship between the feedback and the Sagnac phase shifts depends on the details of the filtering and of the dynamics. In general, a closed-form relationship between the Sagnac and the feedback phase shifts cannot be obtained. A particular case for which a closed-form relationship is achievable is where the filtering is accomplished by a perfect integrator. The differential equation for this case is derived and solved in the following section.

CLOSED-LOOP FIBER GYRO RESPONSE

For purposes of this derivation, the fiber gyro in figure 1 is assumed to be modulated with a square wave and demodulated digitally. The output voltage of the demodulator is shown in Eq. (2).

$$V_{\text{demod}}(t) \approx GP_d \sin(\phi_m)(\phi_s(t) + \phi_f(t)) \quad \text{Eq. (2)}$$

In this expression, G is the electrical gain of the detection circuit, P_d is the detected optical power, ϕ_m is the modulation amplitude, and $\phi_s(t)$ and $\phi_f(t)$ are the time dependent Sagnac and feedback phase

shifts. Since the Sagnac and the feedback phase shifts are nearly equal in magnitude but opposite in sign, the small angle approximation, $\sin(\phi_s(t) + \phi_f(t)) \approx (\phi_s(t) + \phi_f(t))$, is used. This is the input to the feedback loop.

The feedback loop is composed of an amplifier with gain G_{amp} and a filter, which is characterized by an impulse response $h(t)$. The output of the feedback loop is shown below.

$$V_{out}(t) = h(t) \otimes G_{amp} G P_d \sin(\phi_m)(\phi_s(t) + \phi_f(t)) \quad \text{Eq. (3)}$$

In the above expression, the symbol $\otimes$ denotes the convolution operator. The output of the feedback loop is input to the control unit which drives the feedback element. The feedback phase shift is proportional to the control unit output $S_{c,out}$ which in turn is proportion to the control unit input $V_{c,in}$ as shown below.

$$\phi_f(t) = K_f S_{c,out}(t) = K_f K_c V_{c,in}(t) \quad \text{Eq. (4)}$$

Noting that the output voltage of the filter is the input voltage to the control unit yields the following general expression for the feedback phase shift.

$$\phi_f(t) = K_f K_c h(t) \otimes G_{amp} G P_d \sin(\phi_m)(\phi_s(t) + \phi_f(t)) \quad \text{Eq. (5)}$$

The above expression is an integral equation for the feedback phase shift. The integral arises due to the convolution of the filter response with the demodulator output voltage. Closed-form solutions are not possible for arbitrary filters. Arbitrary cases must be handled via numerical techniques. A closed-form solution is achievable for an ideal integrator via Laplace transform techniques. The mathematics of the solution will be skipped and the answer presented below.

$$\frac{d\phi_f(t)}{dt} = -|G_{eq}|(\phi_s(t) + \phi_f(t)) \quad \text{Eq. (6a)}$$

$$|G_{eq}| = K_f K_c K_{int} \otimes G_{amp} G P_d \sin(\phi_m) \quad \text{Eq. (6b)}$$

G_{eq} is an equivalent gain for the closed-loop gyro and is the product of all of the constant factors in Eq. (5). The constant K_{int} is the constant of the ideal integrator. For constant input rotation rates the left-hand side of Eq. (6a) is approximately zero. For this equation to be valid for arbitrary input rates requires that Eq. (1) be satisfied by the feedback phase shift.

As stated earlier, the output of the closed-loop gyro is the control signal $S_{c,out}$. Using Eq. (1), the output signal is expressed in terms of the input rotation rate as given below.

$$S_{c,out}(t) = -\frac{K_{ssf}}{K_f} \Omega(t) \quad \text{Eq. (7)}$$

In this expression K_{SSF} is the Sagnac scale factor and K_f is the proportionality constant between the feedback phase shift and the output signal. Eq. (7) is general in the sense that it is valid for any feedback element when used with an ideal integrator. More of the physics of a closed-loop fiber gyro can be understood if a particular feedback element and control unit are selected and the feedback constant K_f for that combination is derived. The particular case which will be examined is referred to as digital serrodyne². This case is chosen because most major fiber gyro manufacturers in the world are adopting or have adopted a derivative of this technique.

DIGITAL SERRODYNE

In this feedback technique, the feedback element is an integrated optics phase modulator. The key requirement on the phase modulator is that it have a high bandwidth. The phase modulator is driven by a voltage controlled oscillator (VCO) whose output is a digital ramp of constant peak-to-peak amplitude, V_{pp} , and variable frequency, $f(t)$. The output frequency is proportional to the input voltage, $V_{c,in}$, with proportionality constant K_c . The output waveform of the VCO is shown in figure 2.

Note that the duration, T_{step} , of each digital step is nominally equal to the coil transit time, τ . The height of each step, $V_s(t)$, is a function of the frequency and the peak-to-peak amplitude of the digital ramp as given below.

$$V_s(t) = V_{pp} T_{step} f(t) \quad \text{Eq. (8)}$$

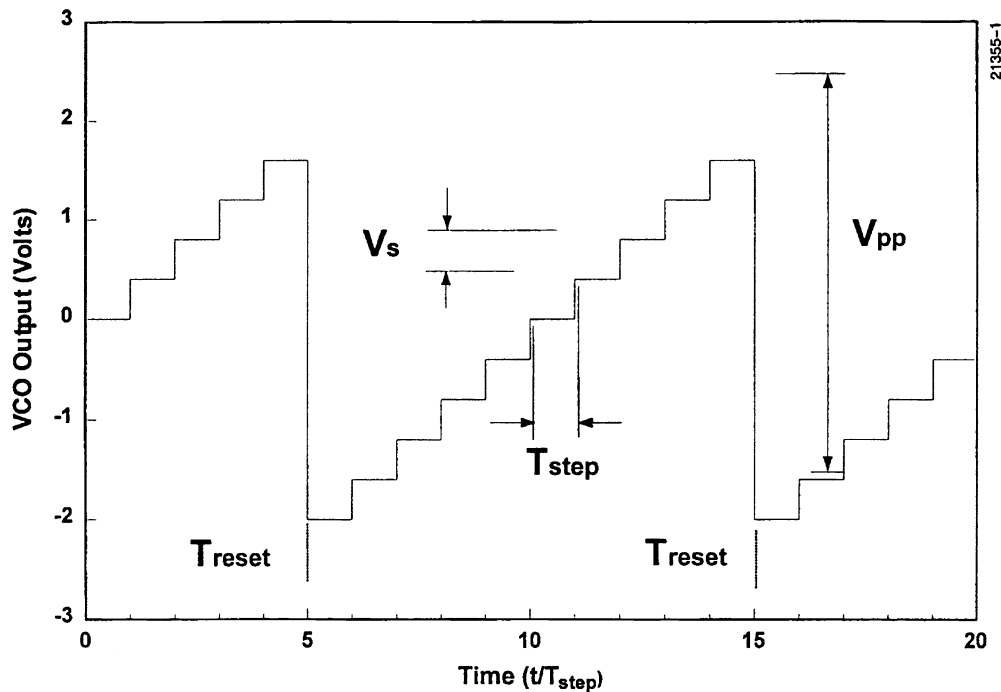


Figure 2. VCO output waveform used in a digital serrodyne closed-loop fiber gyro.

With the above definition, it can be seen that positive frequencies result when the digital waveform is ramping up and negative frequencies occur when the waveform is ramping down. The phase shift induced by a time varying voltage applied to a phase modulator located at one end of the fiber sensing coil is shown below,

$$\phi_f(t) = K_{fp}(V(t) - V(t - \tau)) \quad \text{Eq. (9)}$$

where $V(t)$ is the instantaneous voltage applied to the feedback modulator and K_{fp} is the modulator constant. Referring to figure 2 again, it can be seen that the difference in the instantaneous voltage at two times which are t seconds apart is equal to the step voltage most of the time. Substituting Eq. (8) into Eq. (9) yields the following for the feedback phase shift.

$$\phi_f(t) = K_{fp}V_{pp}T_{step}f(t) \quad \text{Eq. (10)}$$

Comparing the above equation to Eq. (4) and recalling that $S_{c,out}$ is equal to the frequency, $f(t)$, yields the following relation for the feedback constant, K_f .

$$K_f = K_{fp}V_{pp}T_{step} \quad \text{Eq. (11)}$$

Substituting the above into Eq. (7) yields the following expression for the output frequency, $f(t)$, of the closed-loop fiber gyro.

$$f(t) = - \frac{K_{ssf}}{K_{fp}V_{pp}T_{step}} \Omega(t) \quad \text{Eq. (12)}$$

The above expression is the transfer function of the closed-loop fiber gyro using digital serrodyne. The closed-loop scale factor, K_{clsf} , is given below.

$$K_{clsf} = - \frac{2\pi LD}{\lambda c} \times \frac{1}{K_{fp}V_{pp}T_{step}} \quad \text{Eq. (13)}$$

The stability and repeatability of the closed-loop scale factor depend on the stabilities and repeatabilities of several parameters: the length of the fiber sensing coil (L), the diameter of the fiber sensing coil (D), the wavelength of light (λ), the modulator constant (K_{fp}), the peak-to-peak voltage of the digital ramp (V_{pp}), and the duration of the digital step (T_{step}). Each of these parameters is sensitive to temperature. The approximate thermal sensitivities are given in the table in figure 3.

The table shows that the two largest drivers of scale factor stability and repeatability are the modulator constant and the wavelength of the light. It will be seen in the next section that the digital serrodyne technique provides a method for stabilizing the product of the modulator constant and the peak-to-peak ramp voltage by means of a second servo loop.

Parameter	Symbol	Units	Thermal Sensitivity (ppm/°C)
Fiber length	L	m	~0.5 to 5
Fiber sensing coil diameter	D	m	~5 to 20
Wavelength	λ	nm	~300
Modulator constant	K_{fp}	rad/V	~600 to 800
Ramp voltage p-p	V_{pp}	V	~1
Step duration	T_{step}	sec	<1

Figure 3. Closed-loop scale factor parameter thermal sensitivities.

SECOND SERVO LOOP

Figure 2 depicts the VCO output voltage over two cycles. The VCO output voltage resets at times T_{reset} . The phase shift between the CW and CCW optical waves caused by this ramp voltage is given in Eq. (9). The feedback phase shift, ϕ_f , given in Eq. (10), is correct most of the time but is incorrect for a time, T_{step} , after the VCO voltage resets. The correct feedback phase shift at all times is shown below.

$$\phi_f(t) = \begin{cases} K_{fp} V_{pp} T_{step} f(t) & \text{for } T_{reset} + T_{step} < t < T_{reset} \\ K_{fp} V_{pp} (T_{step} f(t) - 1) & \text{for } T_{reset} < t < T_{reset} + T_{step} \end{cases} \quad \text{Eq. (14)}$$

During a time, T_{step} , after the ramp voltage resets a large phase transient between the CW and CCW optical waves occurs. This phase transient is opposite in sign to the phase shift during the ramp. In fact, the sum of the phase shifts in any one cycle of the ramp voltage is identical to zero.

Neglecting the square wave modulation voltage and the Sagnac phase shift due to rotation, the detected optical power during the ramp and during the ramp reset is given below.

$$P(t) = \begin{cases} P_d [1 + \cos(K_{fp} V_{pp} T_{step} f(t))] / 2 & \text{for } T_{reset} + T_{step} < t < T_{reset} \\ P_d [1 + \cos(K_{fp} V_{pp} (T_{step} f(t) - 1))] / 2 & \text{for } T_{reset} < t < T_{reset} + T_{step} \end{cases} \quad \text{Eq. (15)}$$

In order to gain a better physical understanding of Eq. (15), a transfer function plot will be used visualize the above equations. The transfer function plot shown in figure 4 shows the detected power at the ramp reset for three values of $K_{fp} V_{pp}$: $< 2\pi$, $= 2\pi$, $> 2\pi$. It is seen in this figure, that a transient in the detected power occurs if $K_{fp} V_{pp} \neq 2\pi$. This power transient at ramp reset can be used as the input to a servo loop to stabilize the product, $K_{fp} V_{pp}$. Figure 5 depicts a functional schematic of a closed-loop fiber gyro with the above servo loop added. This servo loop is called the second servo loop or, equivalently, the amplitude loop.

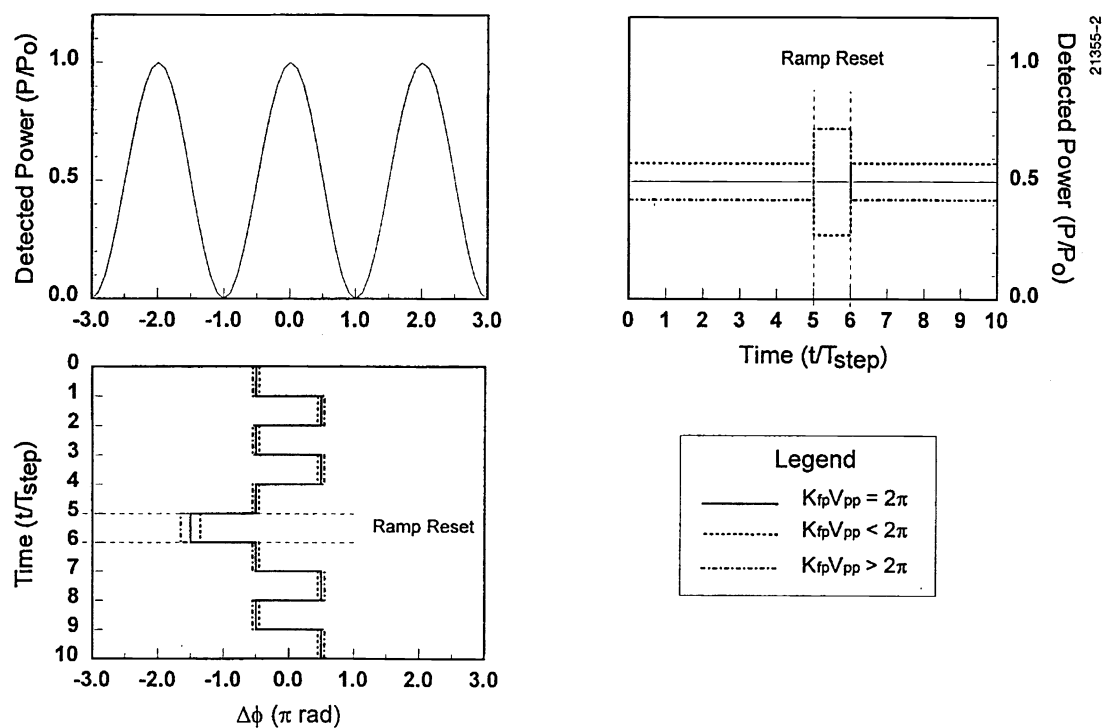


Figure 4. Transfer function plot of a fiber gyro through a reset of the phase ramp. Notice the transient in the phase difference when the ramp resets (lower left curve). The detected power experiences a transient through the reset if $K_{fp}V_{pp} \neq 2\pi$. In this figure only the ramp voltage resets.

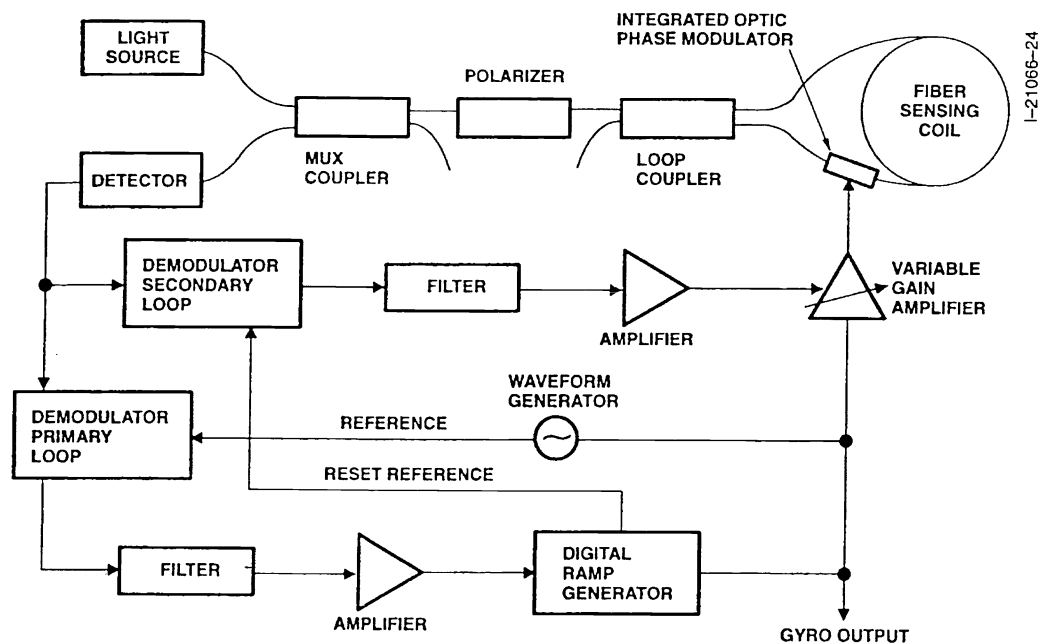


Figure 5. Schematic of a closed-loop fiber gyro employing a second servo loop to stabilize $K_{fp}V_{pp} = 2\pi$. This results in significant scale factor stability improvement.

The servo loop taps a portion of the detected signal from the photodetector prior to rate demodulation and routes it to a second demodulator. This demodulator also has an input from the ramp generator which provides a reset reference to the demodulator. The output from the demodulator is a signal, $S_2(K_{fp}V_{pp},t)$, proportional to the height of the transient. This equation is strictly valid if the square wave modulation and the Sagnac phase shift are neglected. If they are not neglected, then the second factor in Eq. (16) will be modified but the first factor, $\sin(K_{fp}V_{pp}/2)$, will remain unchanged. It is this first factor which is critical for the operation of the second servo loop. The second factor can be viewed as a time-dependent amplitude which affects only the gain of the second servo loop.

$$S_2(K_{fp}V_{pp},t) = P_d \sin \frac{K_{fp}V_{pp}}{2} \times \cos \left[K_{fp}V_{pp}T_{step}f(t) + \frac{\pi - K_{fp}V_{pp}}{2} \right] \quad \text{Eq. (16)}$$

The signal, $S_2(K_{fp}V_{pp},t)$, is plotted versus $K_{fp}V_{pp}$ in figure 6. This signal is filtered and amplified and is then applied to the gain adjustment of a variable gain amplifier. The variable gain amplifier is located between the ramp generator output and the phase modulator. The addition of this variable gain amplifier of gain $G_v(t)$ modifies Eqs. (8) through (16) by replacing V_{pp} with $G_v(t)V_{pp}$. If the gain and filtering of this second servo loop are properly adjusted then the product $K_{fp}G_v(t)V_{pp}$ will be held equal to 2π . The closed-loop, fiber gyro scale factor becomes as shown below.

$$K_{clsf} = -\frac{L}{cT_{step}} \times \frac{D}{\lambda} \quad \text{Eq. (17)}$$

The factor, L/cT_{step} , in the above expression has the approximate value of $1/n$, where n is the index of refraction of the fiber, provided $T_{step} \approx \tau$ (the loop transit time).

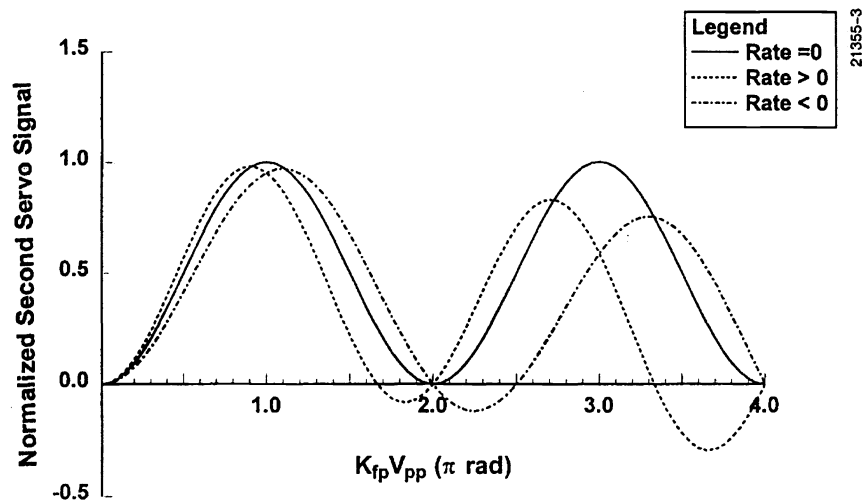


Figure 6. Normalized second servo input signal versus $K_{fp}V_{pp}$ for various input rates.

DEADBAND ERROR⁵

The deadband error is a scale factor linearity error which results from the dynamics of the servo loop. It occurs at low input rates. Figure 7 shows the transfer function of an ideal fiber gyro and a fiber gyro which has the deadband error.

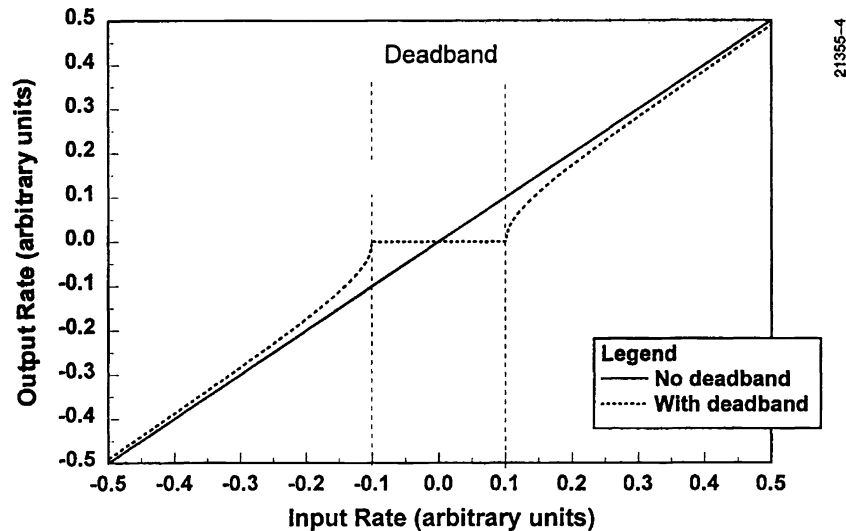


Figure 7. Transfer functions for an ideal fiber gyro (solid line) and for a fiber gyro with deadband (dashed line).

This deadband error in fiber gyros arises from electrical crosstalk in the gyro electronics. The specific crosstalk which causes this effect is that between the phase modulator analog drive and the photodetector analog output. This is shown schematically in figure 8.

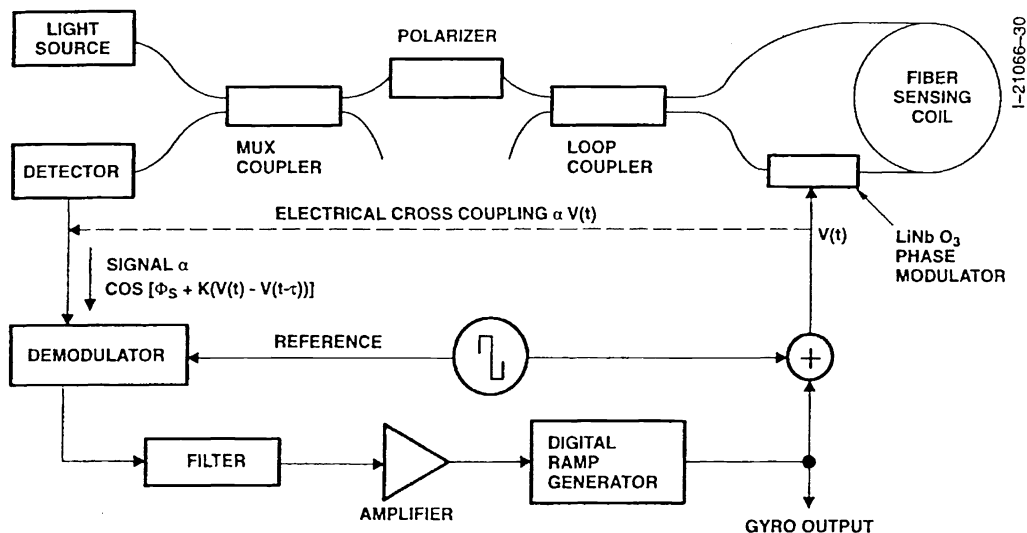


Figure 8. Schematic of a fiber gyro showing the electrical cross-coupling path from the phase modulator drive line to the detector output line.

This crosstalk results in a voltage-dependent bias which is a function of the absolute voltage of the phase modulator drive signal. The servo loop equation for the primary servo loop becomes as shown below.

$$\frac{d\phi_f(t)}{dt} = -|G_{eq}|(\phi_s(t) + \phi_f(t)) - |G_{eq}|\phi_{vdb}(V(t)) \quad \text{Eq. (18)}$$

In this expression, $\phi_{vdb}(V)$ is the equivalent phase bias due to the crosstalk of the phase modulator analog drive signal to the photodetector analog output which is a function of the phase modulator drive voltage.

To continue developing an understanding of the deadband error and ways to circumvent it, it is instructive to consider a specific type of electrical crosstalk. This electrical crosstalk is typical of many closed-loop fiber gyros which use digital serrodyne. Figure 9 shows the voltage applied to the phase modulator versus time. The voltage is the sum of a square wave voltage and a ramp voltage (i.e. $V(t) = V_{\text{ramp}}(t) + V_{\text{mod}}(t)$). The maximum positive voltage is $< 2V$ and the minimum negative voltage is $\geq -2V$. This 4V range is equal to 2π radians of phase shift. The square wave modulation voltage ($V_{\text{mod}}(t)$) has a peak-to-peak amplitude of 1V which corresponds to a $\pm\pi/2$ radian phase modulation. The ramp voltage ($V_{\text{ramp}}(t)$) increases slowly indicative of a non-zero rotation rate.

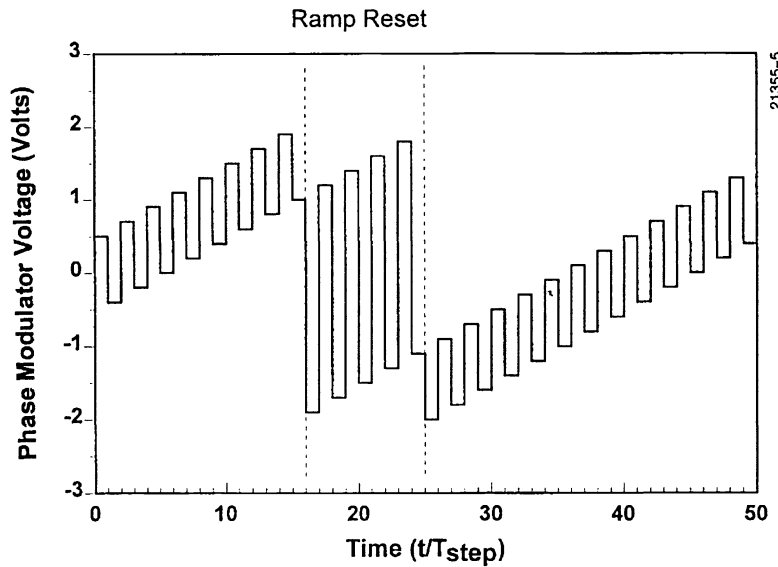


Figure 9. Phase modulator drive voltage which consists of the voltage ramp and the square wave modulation voltage. The combined voltage cannot be less than -2V or greater than or equal to +2V. In this figure both the ramp and modulation voltage are reset unlike figure 4.

Notice that when the ramp voltage, plus or minus the square wave modulation voltage, exceeds the upper or lower voltage limits, the amplitude of the square wave modulation voltage appears to increase by a factor of three to $\pm 3\pi/2$ radians. When the phase modulation is $\pm 3\pi/2$ radians, the modulation voltage is termed to be resetting. This reset occurs for 25% of the ramp period as seen in figure 10.

Examining figure 9, it is seen that the modulation voltage during reset undergoes a 180° phase shift compared to the modulation voltage outside the reset region. The equivalent phase bias error due to the crosstalk of the phase modulation voltage in figure 9 is depicted in figure 10.

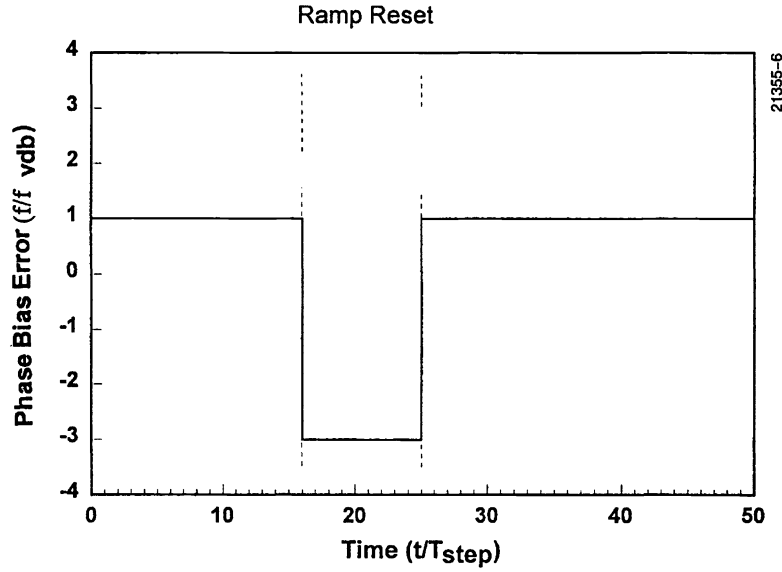


Figure 10. Voltage dependent phase bias error due to electrical crosstalk of the effective modulation waveform shown in figure 9.

At this point, Eq. (18) could be solved. This method does not result in the best physical insight. The quantity, $\phi_f(t)$, is not directly observable. What is directly observable is the ramp voltage, $V_{\text{ramp}}(t)$, applied to the phase modulator. This ramp voltage does not contain any of the modulation voltage. This ramp voltage is given in terms of the feedback phase shift as shown below.

$$V_{\text{ramp}}(t) = \frac{1}{K_f} \int_0^t \phi_f(t') dt' \quad \text{Eq. (19)}$$

Figure 11 shows plots of $V_{\text{ramp}}(t)$ and $\phi_{\text{vdb}}(t)$ for the equivalent phase bias shown in figure 10 and for a large input rate such that $|K_{\text{ssf}} \Omega_{\text{in}}| > |4\phi_{\text{vdb}}|$. This means that the Sagnac phase shift is larger than the equivalent peak-to-peak phase error due to electrical crosstalk. In this case, the voltage ramp decreases and resets but the slope of the voltage ramp varies according to variations in the equivalent phase error.

Figure 12 shows plots of $V_{\text{ramp}}(t)$ and $\phi_{\text{vdb}}(t)$ for the case shown in figure 11 except that the input rate is now small such that $|K_{\text{ssf}} \Omega_{\text{in}}| < |4\phi_{\text{vdb}}|$. This means that the Sagnac phase shift is now smaller than the equivalent peak-to-peak phase error due to electrical crosstalk. In this case, the voltage ramp increases until the first reset is reached then the voltage ramp oscillates about the reset threshold. This oscillation means that the primary or rate servo loop has become unstable due to the presence of a voltage dependent bias. While the primary servo loop is oscillating, the output of the fiber gyro is zero.

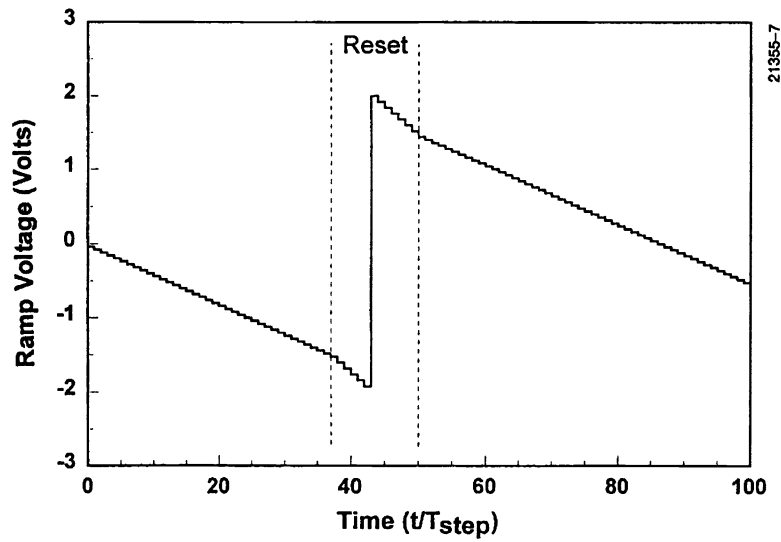


Figure 11. Ramp voltage for the voltage dependent phase bias of figure 10 when $|K_{ssf} \Omega_{in}| > |4\phi_{vdb}|$. The voltage ramps down but the ramp rate depends on the value of the voltage dependent phase bias.

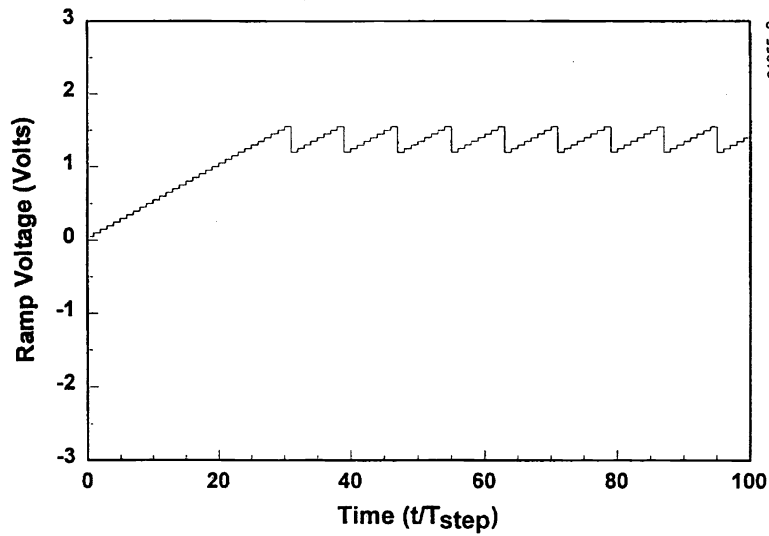


Figure 12. Ramp voltage for the voltage dependent phase bias of figure 10 when $|K_{ssf} \Omega_{in}| < |4\phi_{vdb}|$. The voltage ramps up but it oscillates near the upper reset limit.

DEADBAND COMPENSATION⁶

The deadband error can be eliminated by the addition of a periodic compensation signal, $V_{comp}(t)$, at the input to the integrator as shown in figure 13. The compensation voltage generator also outputs a gate signal which is used to count the cycles of the voltage ramp in a time interval synchronized with an integer multiple of the period, T_{comp} , of the compensating signal.

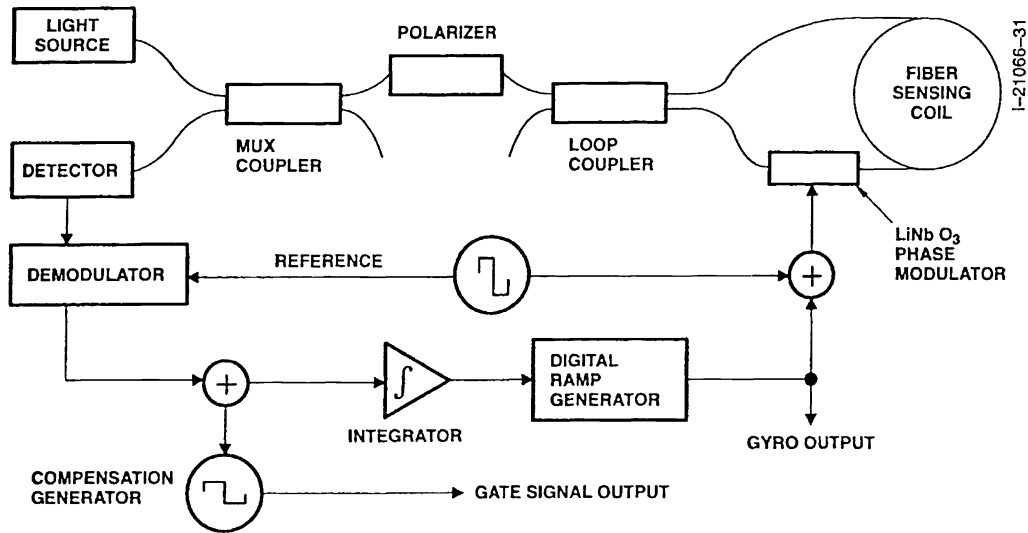


Figure 13. Schematic of a closed-loop fiber gyro with deadband compensation.

The servo loop equation for the primary rate loop is as shown below.

$$\frac{d\phi_f(t)}{dt} = -|G_{eq}|(\phi_s(t) + \phi_f(t)) - |G_{eq}|\phi_{vdb}(V(t)) - |G_{eq}|\phi_{comp}(t) \quad \text{Eq. (20)}$$

$\phi_{comp}(t)$ is the equivalent compensation phase shift caused by the compensation voltage, $V_{comp}(t)$. To suppress the deadband, the amplitude of the equivalent compensation phase shift must satisfy the relations shown in Eq. (21) where the brackets, $\langle \rangle$, denote the time average.

$$\begin{aligned} |\phi_{comp}| &> |\phi_{vdb}| \\ \langle \phi_{comp}(t) \rangle &= 0 \end{aligned} \quad \text{Eq. (21)}$$

The first expression means that the equivalent phase shift produced by the compensation voltage exceeds the equivalent phase shift due to the voltage dependent bias. This results in the right hand side of Eq. (20) not changing sign when the modulator voltage resets and hence the servo loop never breaks into oscillation. This is seen in figure 14, where $V_{ramp}(t)$, $\phi_{vdb}(t)$, and $\phi_{comp}(t)$ are plotted versus time for input rates such that $|K_{ssf}\Omega_{in}| < |4\phi_{vdb}|$.

It is seen that the voltage ramp does not oscillate. The voltage ramp is decreasing when $\phi_{comp}(t) > 0$ and is increasing when $\phi_{comp}(t) < 0$. The slope of the voltage ramp does change as the modulator voltage resets as before.

The second expression in Eq. (21) is necessary to ensure that the compensation voltage adds no bias error of its own to the gyro output. This condition is easily satisfied by capacitively coupling the compensation voltage to the input of the integrator. If the gyro output is averaged over an integer multiple of the compensation period, then the contribution of the compensation voltage to the gyro

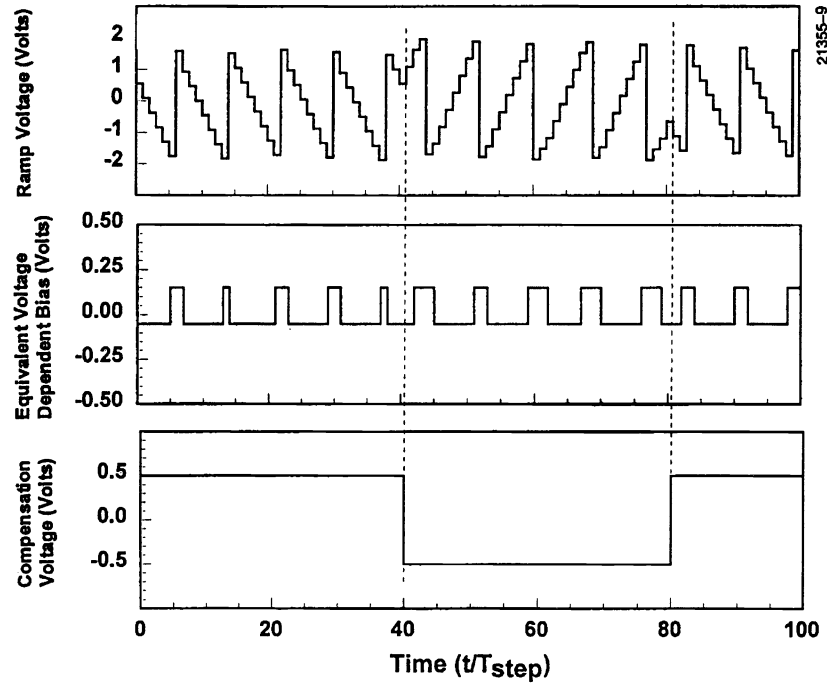


Figure 14. Voltage ramp in the presence of voltage dependent phase bias and the compensation signal when $|K_{ssf} \Omega_{in}| < |4\phi_{vdb}|$. It is seen that the voltage ramp never oscillates.

output is zero. This is better understood by examining the instantaneous frequency, $f(t)$, of the voltage ramp, which can be written as shown below.

$$f(t) = -\frac{L}{cT_{step}} \times \frac{D}{\lambda} \Omega(t) + f_{comp}(t) \quad \text{Eq. (22)}$$

$f_{comp}(t)$ is the contribution of the compensation voltage to the voltage ramp frequency. It has zero mean value over a period, T_{comp} , of the compensating voltage. The number of cycles of the voltage ramp in a time period, T_{comp} , is the integral of Eq. (22) and is given below.

$$\begin{aligned} N_{ramp} &= -\frac{L}{cT_{step}} \times \frac{D}{\lambda} \int_0^{T_{comp}} \Omega(t') dt' + \int_0^{T_{comp}} f_{comp}(t') dt' \\ N_{ramp} &= -\frac{L}{cT_{step}} \times \frac{D}{\lambda} \theta(t) + 0 \end{aligned} \quad \text{Eq. (23)}$$

The second integral in the above expression is zero because $f_{comp}(t)$ has zero mean value over one period. The number of cycles of the voltage ramp over one period of the compensation waveform is proportional to the angle rotated in that time period. Note that the rotation rate, θ , in Eq. (23) is the sum of the actual rotation rate, any rate bias errors described previously, and the voltage-dependent bias. The use of this compensation waveform only eliminates the rate servo loop instability which causes

deadband. It does not eliminate any bias errors, including the voltage dependent bias error due to crosstalk between the phase modulator analog drive line and the photodetector analog output line.

Figure 15 shows the measured output rate versus the input rotation rate for a fiber gyro without and with the deadband compensation described previously. Figure 15a is without the deadband compensation and shows a deadband of 2.7 °/hr. Figure 15b shows the result when deadband compensation is activated.

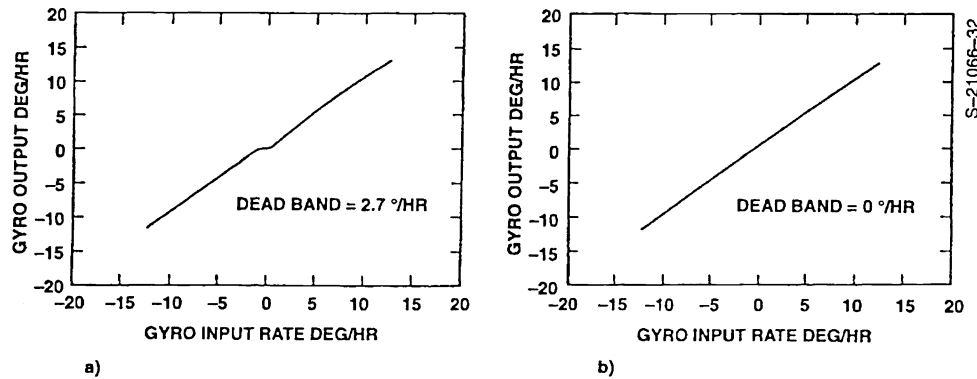


Figure 15. Fiber gyro data showing deadband (a) of 2.7 °/hr without compensation and (b) no deadband with compensation.

SUMMARY

The double closed-loop digital phase ramp (digital serrodyne) and its derivatives have enabled fiber gyros to achieve high scale factor linearity. It has also improved the stability and repeatability of the scale factor. Today, scale factor stability and repeatability are driven by the wavelength stability of the light source and by the dimensional stability of the fiber sensing coil diameter. As with most solutions to a problem, the solution introduces a new problem, deadband error. The deadband error can be eliminated by the use of the compensation signal and synchronized detection technique described previously.

REFERENCES

1. F. Cahill and E. Udd, "Phase-nulling fiber-optic laser gyro," *Optics Letter*, Vol. 4(3), 93-95 (1979).
2. Lefevre, P. H. Graindorge, H. J. Arditty, S. Vatoux, and M. Papuchon, "Double Closed Loop Hybrid Fiber Gyroscope using Digital Phase Ramp," *Proc. 3rd Intl. Conf. Optical Fiber Sensors (OFS '85)*, Technical Digest Series, PDS 7.1-7.8 (1985).
3. A. Pavlath, "The LN200 Fiber Gyro Based Tactical Grade IMU," *Proc. Guidance, Navigation, and Control Conference, AIAA*, 898-904, (1993).
4. A. Patterson, E. L. Goldner, A. Cordova, and D. M. Rozelle, "Test Results over Extended Environments for Litton's Inertial Navigation Grade Interferometric Fiber Optic Gyros," *Proc. Position Location and Navigation Symposium, IEEE*, 176-181 (1994).
5. J. Buschelberger and M. Kemmler, "Closed Loop Fiber Optic Gyro Triad," *Proc. 15th Anniversary Conference: Fiber Optic Gyros, SPIE*, Vol. 1585, 89-97 (1991).
6. Kemmler, P. Grollmann, P. Krasselt, "Suppression of Lock-In in Closed Loop Fibre Optic Gyro," *Proc. 9th Optical Fiber Sensors Conference (OFS-9)*, (1993).